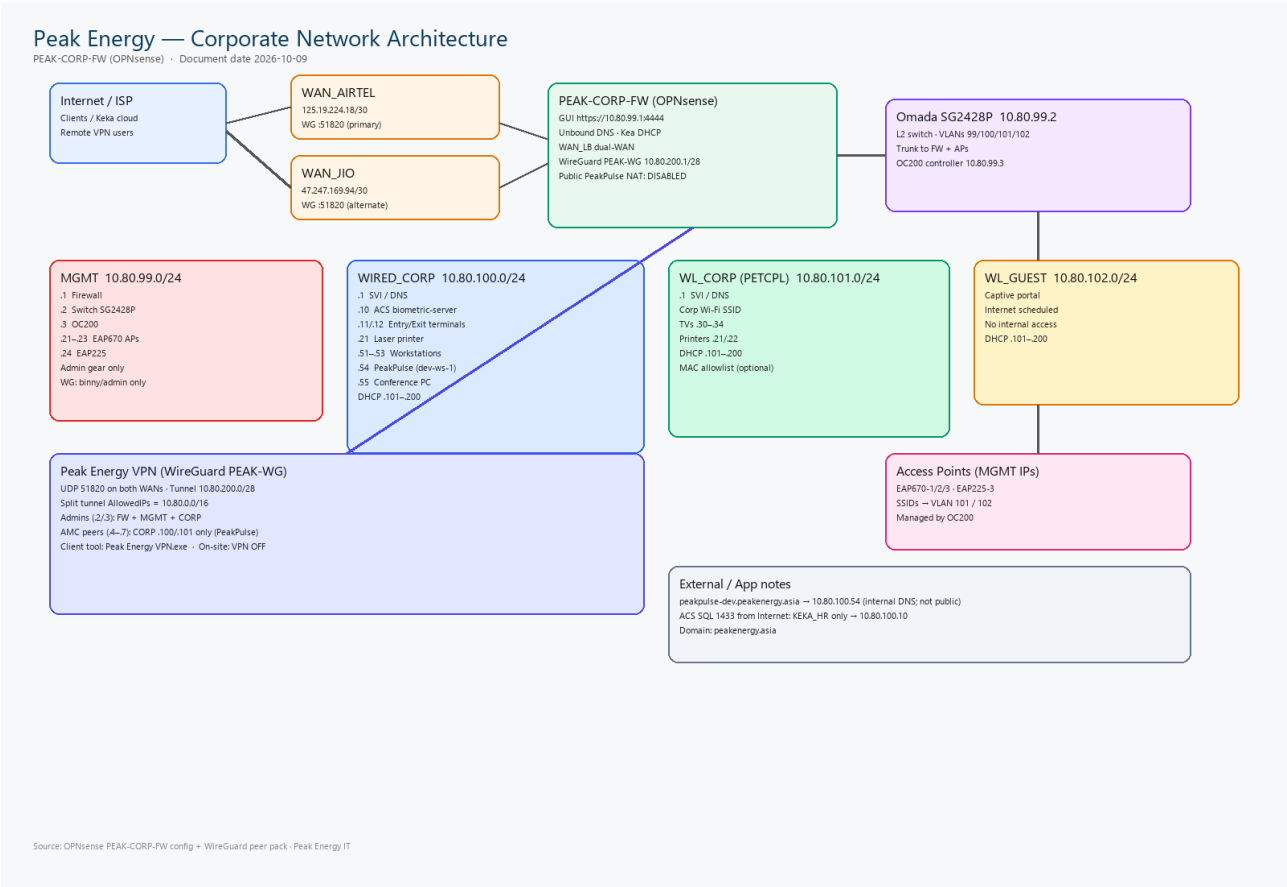


Peak Energy — Corporate Network Architecture

Firewall: **PEAK-CORP-FW** (OPNsense) · Generated 2026-10-09

This document describes VLANs, WAN links, Omada switching/WiFi, DHCP reservations, WireGuard (Peak Energy VPN), and access policy for PeakPulse / ACS.



1. WAN links

WAN	Public IP	Gateway name	Notes
Airtel Fiber	125.19.224.18/30	WAN_AIRTEL	Primary; WG endpoint default
Jio RF	47.247.169.94/30	WAN_JIO	Failover / alternate WG endpoint
WAN_LB	—	WAN_LB	Dual-WAN load balance + failover

2. VLANs / subnets

VLAN / Role	Subnet	Gateway (FW)	DHCP pool	Interface
MGMT (admin gear)	10.80.99.0/24	10.80.99.1	10.80.99.101–200	lan / re0
WIRED_CORP	10.80.100.0/24	10.80.100.1	10.80.100.101–200	opt2 / VLAN 100
WL_CORP (PETCPL)	10.80.101.0/24	10.80.101.1	10.80.101.101–200	opt3 / VLAN 101
WL_GUEST	10.80.102.0/24	10.80.102.1	10.80.102.101–200	opt4 / VLAN 102
WireGuard VPN	10.80.200.0/28	10.80.200.1	static peers	opt6 / wg0

3. MGMT hosts (10.80.99.0/24)

IP	Hostname	Role
10.80.99.1	PEAK-CORP-FW / OpnSense	Firewall GUI :4444, SSH, Unbound DNS
10.80.99.2	sg2428p	Omada switch SG2428P
10.80.99.3	oc200	Omada OC200 controller https://10.80.99.3/
10.80.99.21	eap670-1	Wi-Fi AP EAP670
10.80.99.22	eap670-2	Wi-Fi AP EAP670
10.80.99.23	eap670-3	Wi-Fi AP EAP670
10.80.99.24	eap225-3	Wi-Fi AP EAP225

4. WIRED_CORP hosts (10.80.100.0/24)

IP	Hostname	Role
10.80.100.1	—	WIRED_CORP SVI / DNS
10.80.100.10	biometric-server	ACS / Attendance SQL
10.80.100.11	biometric-entry	Biometric entry terminal
10.80.100.12	biometric-exit	Biometric exit terminal
10.80.100.20	biometric-server-new	ACS new (reservation)
10.80.100.21	laser-printer	HP LaserJet (Ethernet)
10.80.100.51	peak-ws-1	Workstation 1
10.80.100.52	peak-ws-2	Workstation 2
10.80.100.53	peak-ws-3	Workstation 3
10.80.100.54	dev-ws-1	PeakPulse / peakpulse-dev.peakenergy.asia
10.80.100.55	peak801-in-conf	Conference room PC

5. WL_CORP hosts (10.80.101.0/24)

IP	Hostname	Role
10.80.101.1	—	WL_CORP SVI / DNS

10.80.101.21	laser-printer	HP LaserJet (Wi-Fi)
10.80.101.22	printer-md	MD HP Smart Tank Wi-Fi
10.80.101.30	tv-conference	TV Conference Room
10.80.101.31	tv-kalaam	TV Abdul Kalaam
10.80.101.32	tv-ratan	TV Ratan Tata
10.80.101.33	tv-onm	TV Operations & Mgmt
10.80.101.34	tv-md	TV MD

6. WireGuard / Peak Energy VPN

Peer	Tunnel IP	Access
PEAK-WG server	10.80.200.1/28	OPNsense wg0 listen UDP 51820
binny.andrews	10.80.200.2/32	Admin: FW GUI/SSH + MGMT + CORP
admin	10.80.200.3/32	Admin: FW GUI/SSH + MGMT + CORP
jagadeshwar	10.80.200.4/32	AMC: CORP only (.100/.101)
venu.gopal.reddy	10.80.200.5/32	AMC: CORP only
poovarasu	10.80.200.6/32	AMC: CORP only
sanoj.james	10.80.200.7/32	AMC: CORP only

7. Security & access policy

Topic	Current policy
PeakPulse public WAN	DISABLED — no public HTTP/HTTPS port-forward
PeakPulse access	WireGuard or on-site LAN only → 10.80.100.54
WG DNS	Client DNS 10.80.100.1 (Unbound); also listens on WG
WG AllowedIPs	10.80.0.0/16 split tunnel
WG endpoints	125.19.224.18:51820 (Airtel) or 47.247.169.94:51820 (Jio)
MGMT via VPN	Only WG_FW_ADMINS (binny + admin)
FW GUI via VPN	Only WG_FW_ADMINS → (self) :4444 / SSH
AMC peers via VPN	CORP .100 + .101 only (PeakPulse etc.)
ACS SQL from Internet	KEKA_HR only → 10.80.100.10:1433
Guest Wi-Fi	VLAN 102 + captive portal; scheduled internet Mon–Fri 07:00–19:00
On-site + WG tip	On PETCPL turn VPN OFF (Peak Energy VPN.exe); remote turn ON

8. Logical traffic paths

On-site (PETCPL / wired): Clients use VLAN gateway directly. Peak Energy VPN should be **OFF** so 10.80.x is not forced into the tunnel.

Remote: Peak Energy VPN **ON** → UDP 51820 → PEAK-WG → CORP (.100/.101). Admins also reach MGMT (.99) and firewall GUI.

Guest: VLAN 102 captive portal; blocked from MGMT/CORP; internet by schedule.

Confidential — Peak Energy IT. Derived from OPNsense config-PEAK-CORP-FW and WireGuard peer pack.